

A System Dynamics Model for Managing Service Desk Capacity

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Abstract

Information technology (IT) services have become more complex and diversified. The service desk establishes a single channel of communication between the internal IT provider and its users, through which incidents and problems are analyzed, escalated and resolved. Managers need to analyze different operating scenarios regarding service desk capacity to make effective decisions about what should be improved to meet the business demands. This paper presents a system dynamics model for evaluating an organization's service desk's service capacity. The model allows the performance simulation of scenarios, estimating the service capacity. Preliminary results are promising and seem to contribute to the improvement of capacity management processes.

Keywords: IT service management; Service desk; System dynamics.

I. INTRODUCTION

The growing importance of information technology (IT) for companies has made business dependent more and more on IT services [1]. It is of interest to manage and maintain IT services effectively [2]. The service desk is the main communication channel between the IT department and the users of its services. When IT service users face problems, they usually call the service desk in order to obtain a solution in the most objective way possible. Important considerations concerning the performance of customer service include time of service, waiting delays and call registration, among others. Other indicators that should also be considered during the regular monitoring of service are the call dropout rate, the technical knowledge of the subject by the support staff, requests answered within and outside the support time slot (formalized in service level agreements - SLAs) [16], incident monitoring, among other aspects.

Considering the large amount of services offered in the daily life of an IT area, it is possible to make the following questions: What is the current capacity of the organization's service desk? What is the demand of essential services to the organization? What is the best service desk configuration in order to meet the current demand?

This paper proposes a new model, to estimate the staffing capacity to be allocated to the service desk so that it meets the organization's business demands. The simulation results were useful for managers to assess and adjust the service capacity of the IT area to meet real business demands.

II. LITERATURE REVIEW AND RELATED WORK

At a high conceptual level, several other studies in Business-driven IT management (BDIM) [3, 4] relate to ours.

The lack of a definitive and consolidated understanding of what constitutes the IT capabilities of an organization and how to measure this ability brings great possibilities for many research papers [15]. Authors of [5] identified which IT skills are essential for business success, and named them "core capabilities". The IT capability is characterized as the organizational ability to mobilize and apply IT resources in an integrated way or the co-presence of other resources and organization skills [7]. The IT Capability Maturity Framework (IT-CMF) [6] uses the maturity level approach to enhance IT capabilities of organizations over time. The IT-CMF supports oriented IT management to create value.

The work in [8] proposes an IT services reporting framework in order to help organizations to implement process IT services improvement.

Service Desk

The service desk is the first contact point between users and the IT area when there is any need of support [9]. In service desk, all requests, calls and events must be logged [10]. Often the staff (service capacity) of the service desk is not enough to bear the needs of its users in times of peak or because of certain problems or unexpected incidents. What to do then do to keep IT risk at an acceptable level? [16]. Frequently, to estimate a structure that can meet these unexpected events on an emergency basis, managers use their own experience to decide what should be improved.

System dynamics

System dynamics (SD) was presented in [11], in order to model and analyze the behavior in complex scenarios of social systems. The technique is used in the modeling of various types of systems: social, economic and environmental, where a holistic view has important role and feedback loops are essential for understanding the inter-relationships [12].

SD is based on a direct structure of flows and stocks designed to model systems with many variables and delayed feedback between variables. Figure 1 shows an example of SD modeling applied the process of receiving requests by an IT service desk. SD foresees the use of two types of models: influence diagrams or causal-links (qualitative models) and models based on computer or stock-flow diagrams (quantitative). The first is considered the heart of the SD. Influence diagrams are sufficient for identifying the various feedback loops (strengthening and balancing) and disruptive factors of the system under modeling. However, only through computer simulations (quantitative models) one may have a picture of the actual dynamic effects of these feedback loops and a greater understanding of the system under analysis. This encourages exploration of experimentation in SD [13] [14].

System dynamics models feature a dynamic view of the performance of business systems [14]. The SD models are more used in modeling a full or a line of business organization, and little used for low-level workflow models.

A model that uses system dynamics is not static, for it represents how a change in variables affects a process [14]. The authors of [17] proposed a system dynamics model based on the concepts of dynamic capabilities and benefits management for evaluating and managing customers' IT capabilities that need to be applied to outsourcing relationships over time.

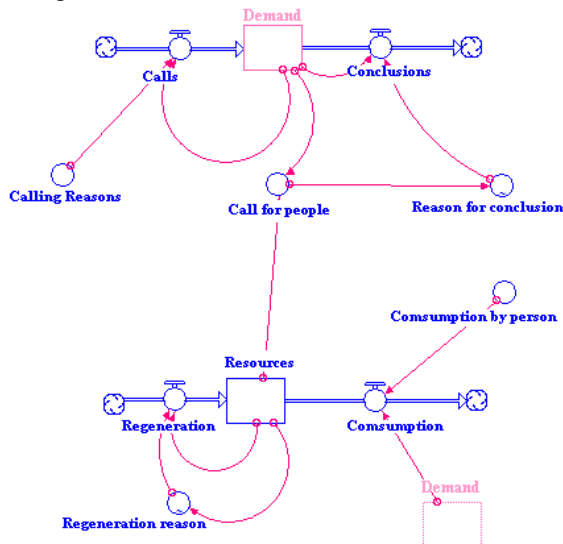


Figure 1. Modeling with SD for an IT service desk

III. THE PROPOSED MODEL

The service desk features vary according to the company size, business criticality and type of industry.

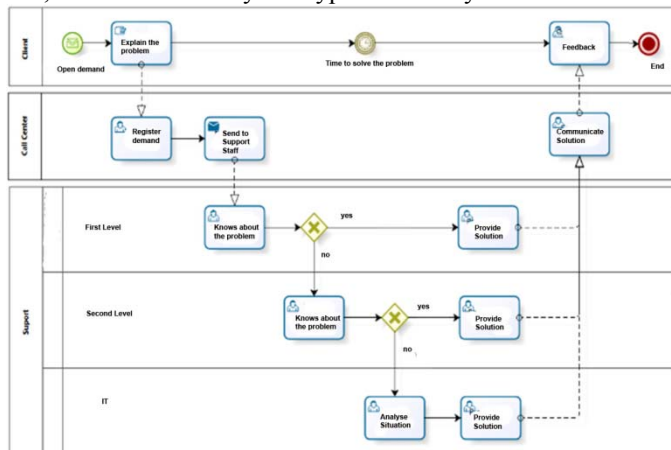


Figure 2. Proposed organization of a service desk

The service desk follows a process that begins with the opening call, made by the client. In the first stage, the caller is helped by the first level, which will attempt to solve the problem in service support slot time. If no solution is found at this level, the call can be routed to the second level and third level (specialized IT professionals).

The proposed model starts with a help request from the user. The service steps, were modeled based on Business Process Model and Notation (BPMN), as shown in Figure 2.

IV. CASE STUDY AND RESULT ANALYSIS

We proceeded a case study at a Brazilian company. For confidentiality, we will call this company "Alpha". Alpha

company has a staff of over 800 employees performing specific IT jobs, 600 work in the service desk, handling about 4,000 daily calls.

The case study involved the following steps: 1. Literature review; 2. Model development; 3. Meeting with the managers; 4. Data collection; 5. Using the model indifferent scenarios; 6. Presentation and discussion of results with managers; 7. Tabulation results; 8. Model validation; 9. Disclosure of the final results.

Simulation Model

The quantitative evaluation of service desk capabilities, seeking to optimize the allocation of resources in order to achieve the planned results is still a challenging problem, especially when it involves the skills of people and the impact of the tools and techniques used to carry out the work. Among the restrictive factors, we can cite the loss of internal expertise, often generated by high turnover is a major risk factor for organizations that rely on IT.

Model Objectives

The modeled resources takes into account the staff availability that the service desk has to meet the organization's demands. The model seeks to evaluate the required capacity to meet service demand because adding more resources (staff) is not always the solution. The objectives can be achieved considering other parameters such as time, season of the year, adopted technology, staff know-how. Due to space restrictions, this article identifies the risks but does not discuss ways to mitigate them. Our model was segmented into "views", reflecting the organization of policies captured in the modeling phase. Aspects such as customer seasonality and absenteeism were taken into consideration: depending on the time of year, there will be the need for more staff or "service points" (SPs) in the service desk organization.

Input and output data

The model has parameters divided into four categories: internal input agents, external input agents, assessment and output results. The parameters related to internal and external input agents characterize the services' demands, the IT resources that an organization has and the complexities of IT demands. Evaluation parameters are used to adjust the model's behavior for the scenarios being simulated. They also represent intermediate information, obtained from the inputs, by providing feedback on the results. The output parameters are directly proportional to information arising from cause-effect relationships involving the model's input parameters. Our model presents results for the expected performance of IT resources (in terms of cost, quality and resource consumption) in relation to the input data.

Model results

Results for three simulations based on data from the alpha company, in which we consider factors related to the amount of service desk staff, number of calls and staff skill level. The first two items are quantitative criteria; the latter is qualitative, where the manager must assess the strengths and weaknesses of each team member.

Incident and Problem Management

In service desk management, there are various demands and depending on the source of demand, one can decide whether the service will be performed by the team of the first, second or third level. IT demands considered in the proposed model may be seen as consisting of two parts: the part related

to the incidents, which receives allocation of resources from the first and second service desk levels because they relate to problems of small or medium complexity; and, the part related to services in situations where resources with expertise in IT (level 3) are required. In this scenario (see Figure 3), the user can simulate the amount of demand for the service desk.

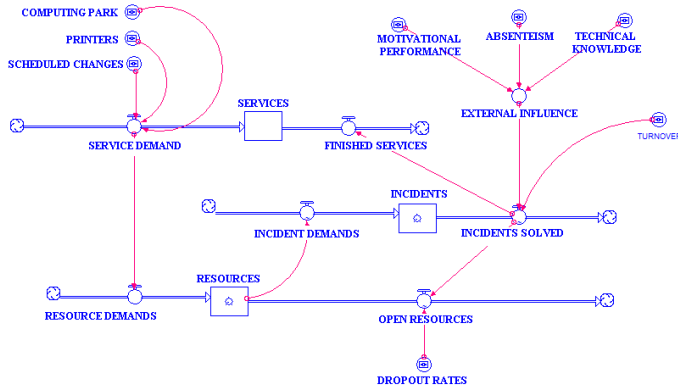


Figure 3. Service desk scenario simulation

Incident Leads

The incidents' demands in the organization are evaluated from output metrics Calculation of Percentage Calls (CPC) and Technical Calculation versus Demand (TCD). It was considered in the research that capacity consists of a productivity rate - that is, it could be represented by the metric Total Calls Within Working Hours of service desk (TCWWH). There is also the Total Calls Served Within Working Hours in the service desk (TCSWWH). From these two metrics, you can calculate the *Percentage of Calls Served Within Working Hours* in the service desk (PCSWWH) by equation:

$$PCSWWH = \frac{TCSWWH}{TCWWH} * 100 \quad (1)$$

Servicing the calls initiates during office hours of the company's service desk. If the service desk functions from 8 a.m. to 6 p.m. and a given call is registered at 7:30 p.m., this will only begin to be monitored at 8a.m. the next day.

Service demands

The demands are serviced by IT experts. We assume that the working capacity of an IT expert is measured by a rate of productivity - ability to perform activity in a time interval. We represent it by two metrics: Window for Running Demand (WRD) and Average Time of Implementation of Activity (ATIA). Both metrics are expressed in minutes. From these two pieces of information it is possible to measure the amount of experts required to perform an activity within a time interval. The equation below shows the *Calculation of Technical versus Demand* (CTD):

$$CTD = \frac{WRD}{ATIA} \quad (2)$$

Service Level Agreement

In addition to the service and solution of calls, there is a requirement that solutions be provided with an expected level of performance, as specified in a SLA (Service Level Agreement). An assigned group of IT experts is responsible for each IT service. We consider that, for each demand, there is a Window for Running Demand (WRD) and Average Time of Implementation of Activity (ATIA), and both are evaluated in terms of minutes.

Through *Total of Calls Open Within Working Hours* (TCOWWH) of the service desk on the day and the *Total of Calls Served Within a Specified Time* (TCSWST), the manager can identify if the service is meeting the SLA that has been

agreed with the customer. The percentage of SLA that is being met is given as:

$$SLA(\%) = \frac{TCOWWH}{TCSWST} * 100 \quad (3)$$

Work Scope

We considered the service demand in the last 6 months. The objective was to evaluate the ability to meet the demands at times of peak demand and at the more stable ones. For the research, we had access to logs of serviced calls, levels of demand, average troubleshooting time and the applicable SLAs.

Performance Impacts

Besides all the considerations that *Alpha* uses for its calculations and predictions, the proposed model also considered the level of service desk staff absenteeism that is, the level of lack of resources (LLR), who impact on performance can be very high. The research also considered the *Motivational Performance Level* (MPL) and the *Technical Knowledge Level* (TKL). Some particular events may interfere with the performance of resources (staff), generating absence from work, punctuality problems or early departure. It is important to consider absenteeism because if it were to happen in mass, it would interfere with the calculation of the SLA, CTD and PCSWWH. When motivated, staff is more productive, with better performance and the more it understands the activities of the organization, the faster it will work, and the better its quality. These factors were taken into account. Once the LLR, MPL and TKL are known, one can calculate the External Influence Factor (EIF) by equation:

$$EIF = \frac{LLR + MPL + TKL}{3} \quad (4)$$

This information is qualitative and its values may vary, depending on the time of the year.

Simulation experiments

To develop our proposed model, we used the SD tool *Stella*. For the simulations, it was considered that the service desk has a minimum daily demand of 4,000 incidents and 10 services, and a team of 600 SPs. The information was obtained from the logs of *Alfa*.

A. First simulation

Whereas the number of SPs and demands of the incident type and service remain unchanged, were evaluated grade 0 to LLR, 7 to TKL, and 5 for the MPL. Figure 4 presents the graph showing the result. For this scenario, *Alpha* company can have a balanced service to the support demand for IT services, running at its capacity limit, but without overloading the team.

B. Second simulation

In the second simulation, we maintained the number of SPs and the demands were for 7000 to service demands. Grade 2 was used to LLR 7 for TKL, and increased the MPL to 7 and a turnover of 10%. Figure 5 shows the graph of the simulation results. The scenario presents a drop in performance of the service by part of the staff. Whereas we had turnover, LLR and service demands, the MPL increase had considerable influence in reducing the impact.

C. Third simulation

The third simulation considers a peak period of services. The service desk has 700 SPs, 10,000 calls per day. There is still a 10% turnover one, grade 2 LLR and for both TKL and MPL were grade 8. As shown in Figure 6, we can note that this scenario shows a balance between service demand and the level of care that resources are having. An increase in service

performance by a part of the staff because of improvements in TKL and MPL.

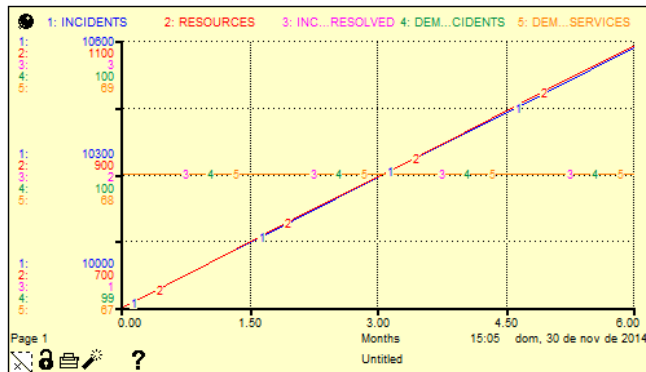


Figure 4. First scenario's simulation.

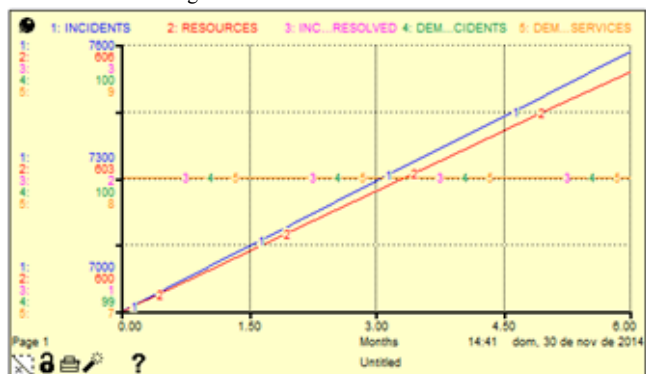


Figure 5. Second scenario simulation

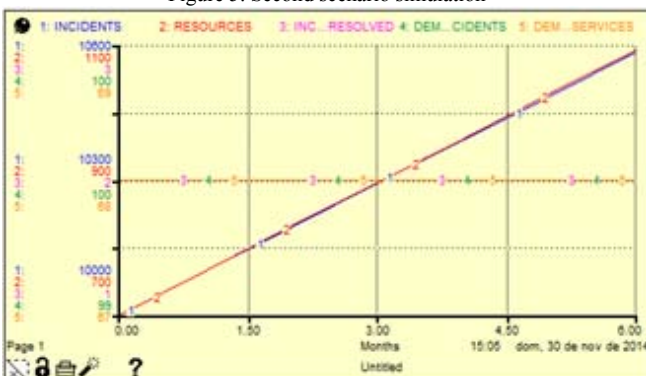


Figure 6. Third scenario simulation

D. Analysis of the results and managers considerations.

During the survey phase of the research, we noticed that managers had difficulty estimating the knowledge level of the service desk staff. This is due to the fact that besides evaluating the staff technical capacity, which is more easily captured from its work performance, there are the qualitative criteria such as motivation and knowledge, and external factors such as turnover and LLR that need to be into account.

Results indicated that these more subjective criteria can make a difference. The motivational level can be identified through feedback from employees and the Human Resources department. The level of absenteeism, which varies from region to region in the country, can be predicted in advance considering the festive events and the history of previous years. The degree of turnover can be estimated by the stress generated by the function. As a person gains experience, he or she will also to move on, quitting from the service desk.

V. FINAL CONSIDERATIONS AND FUTURE WORK

This paper presented a model based on system dynamics to support the decision making process in the management of the capability of a service desk.

The main conclusion from using the model is that increasing the number of staff in the service desk function may not be a sufficient measure to improve service capability. This can be noted from the simulation results where the same number of people working on service calls can lead to very different results when there is a higher level of knowledge and motivation. These two items of qualitative evaluation will influence considerably in the ability to service calls.

Validating the proposed model is a multi-year effort. Here we have just started the validation process, through a preliminary assessment at a Brazilian company specialized in IT service desk outsourcing. By means of an explicit approach of interviews and questionnaires with professionals who are well informed about the real system, we sought to identify whether the proposal seemed reasonable in its initial predictions and performance evaluation. Endorsement from these professionals is encouraging.

In future work we intend to improve and evaluate our model for a longer period of time in the service desk of the alpha company and of other companies, including those that run its own, internal IT help desk.

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